

Arjun Sidhu

sidhuarjunvir@gmail.com | linkedin.com/in/arjunvir-sidhu | github.com/rjnvr | +1(437)770-9365 | arjunvirsidhu.com

PROFESSIONAL EXPERIENCE

Automation & Solutions Engineer (Co-op)

Jan 2025 - Aug 2025

Ministry of Public and Business Service Delivery and Procurement (MPBSDP)

Toronto, ON

- Reduced manual processing time by ~45% by automating document approval process using Power Platform
- Built internal platforms used by 15+ stakeholders to centralize operations & reporting workflows
- Led client-facing sessions to scope automation solutions, translating business needs into technical implementations
- Built production-grade automation systems (Power Apps, Automate, SharePoint) for workflow orchestration, CRM, & call tracking

Automation & Solutions Engineer (Co-op)

May 2024 - Aug 2024

Ontario Ministry of Transportation (MTO)

Toronto, ON

- Decreased manual processing time 50% by automating internal approval workflows
- Built data collection and approval routing automations, improving data consistency and management processes across teams
- Owned end-to-end delivery of automation solutions, from requirements to deployment in a cross-functional environment

Jr. Technical Analyst (Co-op)

Sept 2023 - Dec 2023

Ontario Ministry of Transportation (MTO)

Toronto, ON

- Developed a Streamlit-based AI assistant for automated data retrieval and analysis from structured datasets
- Built a text-to-SQL system using Azure OpenAI, enabling non-technical users to query PostgreSQL databases via natural language
- Improved response accuracy and reduced API cost by ~20% through clever prompt engineering and optimization
- Secured project approval and funding by presenting technical solutions to stakeholders & created technical documentation for team

PROJECTS

ChangeFlow Platform | React.js, TypeScript, Node.js, Claude API, AWS S3 (2026)

- Built a production-ready AI SaaS platform for managing construction change orders and workflows
- Developed an agentic AI system using Claude API that analyzes documents, extracts risks, and autonomously triggers actions (task creation, alerts)
- Designed a RAG pipeline (chunking + embeddings + vector search) enabling context-aware querying across project documents
- Implemented multi-step LLM orchestration with tool execution and memory, simulating real-world autonomous workflows
- Architected full-stack system (React frontend, Node backend, AWS storage) supporting document ingestion and real-time interaction

AgentEval Harness | FastAPI, React, TypeScript, SQLite, Anthropic API, Python (2026)

- Built an agent evaluation system for RAG + tool-using AI workflows, testing across 20+ cases with metrics for answer accuracy, tool correctness, latency, cost, & failure modes
- Designed scoring & failure analysis pipeline (hallucination detection, action validation, retrieval inspection) to diagnose agent reliability issues
- Developed a full-stack evaluation dashboard (FastAPI + React) to run experiments & analyze agent performance

EDUCATION

Bachelor of Computing Honours, Computer Science (Co-op)

Sept. 2021 – Apr. 2026

University of Guelph

Guelph, ON

- Relevant Courses: Discrete Structures 1 & 2, Computer Networks, Software Engineering, Parallel Programming, Operating Systems, Computational Intelligence, Theory of Computation, Compilers, Computer Graphics

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, Java, SQL, TypeScript

AI/ML: RAG, LLM orchestration, prompt engineering, agentic workflows

Frameworks & Libraries: Node.js, React.js, Streamlit, Pandas, NumPy, Axios, PyTorch, Scikit-Learn

Tools & Platforms: Git, AWS (S3), Azure OpenAI, Anthropic API, PostgreSQL, Azure DevOps